Data Management Project - AY 2025-2026

2025 F1 SEASON DATABASE

Design and implementation of a Database of the 2025 Formula 1 Season

Authors:
Anna Rebecca Sala - MAT. 894569
Giorgia Brioschi - MAT. 894407

CONTENTS

INTRODUCTION 2
1. DATA ACQUISITION 3
1.1 Web API 3
1.2 Web Scraping 4
2. DATA PREPARATION 5
2.1 Data Cleaning 5
2.2 Data Integration and Enrichment 6
2.3 Data Quality 7
3. DATA STORAGE 8
3.1 Database 8
3.2 Queries 8
CONCLUSIONS 11

INTRODUCTION

The aim of this project, which falls within the field of data management, is to consolidate the 2025 Formula 1 season into a single and comprehensive database, named *f1_2025*. This database provides users with a unified, clear, complete and interconnected source through which they can explore detailed information and characteristics related to the most recent F1 season. Specifically, the database contains information about races, results, drivers, constructors, circuits and final standings.

The idea of creating this database arose from the desire to answer some research questions that investigate, for example, performance trends in relation to the type of circuit, the grid position and a summary of a driver characteristics and performance. To answer those research questions, it is therefore necessary to have a database that contained sufficient and detailed information about all the aspects related to the F1 season.

This report documents the different phases of the project by following the data science pipeline that describes the data lifecycle. In particular, the pipeline is implemented through Jupyter Notebooks in Python language and includes the following steps:

1. Data acquisition: collection of data from multiple sources using Web APIs and Web Scraping methodologies, with raw data stored as CSV files.
2. Data preparation: evaluation of data quality and application of data cleaning and preprocessing techniques to make the data suitable for further processing. Automated integration and enrichment of heterogeneous data sources in order to achieve a unified and coherent view.
3. Data storage: design and implementation of a database model for storing the processed data. In this project, MongoDB, a document-oriented NoSQL database, was selected. The database is queried through a set of representative queries using Python and the PyMongo library.

All operations performed across the various phases are accessible and reproducible using the files available in the shared Drive folder, with the support of the operational guide.

1. DATA ACQUISITION

After defining the objectives of the analysis, the project proceeded with the data acquisition phase, which involves collecting and importing data to be used in subsequent stages of the workflow. In this project, the datasets were obtained using two complementary methodologies: Web API and Web Scraping. These approaches were adopted due to the constraints of the selected web resources, which did not allow manual data downloads. The overall strategy was to select a web page to enrich and complement the primary API-based data source with additional information.

1.1 Web API

An Application Programming Interface (API) is a set of standardised protocols and tools that enables communication between different software systems. APIs allow applications to exchange data and perform specific operations, such as retrieving information from third-party providers. The data returned by APIs are often formatted as JSON documents, structured into nested objects and arrays. Interaction with an API typically occurs through endpoints and filtering parameters, which allow users to select and manipulate specific subsets of data.

In this project, the Jolpica F1 API was employed. This open-source API, maintained by a small group of volunteers, provides access to a wide range of Formula 1 data. It was selected due to its high level of completeness and its coverage of key entities within the Formula 1 ecosystem, including seasons, races, drivers, constructors, circuits and standings.

Requests were sent to the base endpoint *https://api.jolpi.ca/ergast/f1/*, filtering the results for the 2025 season. The following endpoints were queried:

- */2025/results*: provides races results for the 2025 season. From this endpoint, only non-redundant attributes were retained, including the round number and, for each classified position, points scored, driver ID, constructor ID, grid position, number of laps completed, final race status, total race time and fastest lap time.
- */2025/sprints*: contains sprint races results for the 2025 season. The same set of attributes collected from the race results endpoint was extracted.
- */2025/drivers*: includes information about drivers who participated in the 2025 season. For each driver, the following attributes were collected: driver ID, permanent number, first name, last name and date of birth.
- */2025/constructors:* includes information about constructors who participated in the 2025 season. For each constructor, the following attributes were collected: constructor ID and name.
- */2025/races*: provides detailed information about each race in the championship, including the round number, race name, circuit ID and the date and time of both the main race and the sprint event, when applicable.

From all API sources, the season column was not included, as it always contains 2025. The url column was omitted as irrelevant for the project's goal and any other columns duplicating information from the scraped datasets were also excluded.

1.2 Web Scraping

Web scraping techniques were used to extract data from web pages that do not offer alternative programmatic access methods. A web scraper is a software tool that automates the process of accessing a website, parsing its content and extracting structured information.

Wikipedia was selected as the scraping data source, as it hosts multiple Formula 1–related pages containing detailed and well-structured information, often presented in tabular form.

The following Wikipedia pages were scraped:

- "*List_of_Formula_One_circuits*": the full table containing circuit-related information was extracted, including circuit name, map, type, direction, location, country, last length used, number of turns, associated Grand Prix, seasons and number of Grands Prix held.
- "*List_of_Formula_One_drivers*": a single table was extracted. Records were filtered to retain only those drivers who competed during the 2025 season. The collected attributes included full name, nationality, seasons competed, championships, race entries, race starts, pole positions, race wins, podium finishes, fastest laps and total points.
- "*List_of_Formula_One_constructors*": two tables were extracted and concatenated: one listing constructors active in the 2026 season and another containing the remaining ones. These tables include information about constructors such as name, engine supplier, licensed country, base location, seasons competed, races entered, races started, number of drivers, total entries, wins, points, pole positions, fastest laps, podiums, World Constructors' Championships (WCC) and World Drivers' Championships (WDC), antecedent teams.

After extraction, all the files contaoning the obtain data were stored as CSV files. This enabled subsequent data cleaning and preprocessing operations using the pandas library, ensuring data consistency, clarity and the removal of redundancy prior to database insertion. The following CSV files were produced during this phase: *circuits_wiki.csv*, *constructors_api.csv*, *constructors_wiki.csv*, *drivers_api.csv*, *drivers_wiki.csv*, *races_api.csv*, *results_api.csv* and *sprint_api.csv*.

2. DATA PREPARATION

The data preparation phase aimed to improve the quality of the data through data cleaning and integration operations. This step allowed the identification of formatting issues, inconsistencies and structural missing values, it also allowed to unify and enrich heterogeneous datasets obtained from the data acquisition step.

2.1 Data Cleaning

The main data cleaning and preprocessing operations focused on column formatting and the removal or handling of unnecessary and inconsistent data.

With respect to missing values and data removal, several actions were performed.

- First, records corresponding to drivers with missing data were removed from the *drivers_api* dataset. A manual verification showed that these drivers were included because they participated in free practice sessions, but did not actually compete in any race during the 2025 season.
- In the *circuits_wiki* dataset, the “maps” column was removed, as the scraping process did not allow the associated circuit map images to be downloaded.
- Similarly, in the *constructors_wiki* dataset, the “engine”, “based in” and “antecedent teams" columns were dropped, since they were available only in one of the two scraped tables and would therefore introduce inconsistencies.
- Finally, the races dataset contains missing values in the column related to sprint race date and time. These missing values do not represent errors, but rather a structural characteristic of the data, as they indicate that no sprint race was held for the corresponding event. Additionally, the results and sprint datasets contain missing values in the columns related to race time and fastest lap. These do not represent data errors, instead they indicate cases where a driver did not finish the race or sprint event.

Regarding data formatting and transformation, additional preprocessing steps were applied.

- In the *circuits_wiki* dataset, the column representing circuit length was modified to keep only the length expressed in kilometres, facilitating its use during query execution. Moreover, the direction column was standardised to contain only three predefined values.
- The seasons column present in the *circuits_wiki*, *drivers_wiki* and *constructors_wiki* datasets was originally represented as year ranges. This column was transformed using an expand operation to produce an explicit list of seasons in which each entity participated, thereby improving readability and consistency.
- Several textual fields in the *circuits_wiki*, *drivers_wiki* and *constructors_wiki* datasets, such as names columns, contained additional special characters (e.g., “*”, “|”, “!”). These characters were removed to ensure consistency and proper matching across datasets. Similarly, in the races dataset, the “Z” character was removed from the time column.
- In the constructors_wiki dataset, when multiple team names were associated with a single record, only one standardised name was retained. Moreover, when multiple countries were

associated with a single constructor, this information was transformed using an expand operation to produce an explicit list of countries.

- In the drivers_wiki dataset, for each record, the championship information was modified by keeping only the number of titles, removing any additional notes.

Moreover, data types were checked across all files, ensuring consistency for numeric, text and datetime columns.

Finally, to ensure alignment with the objectives of the project, the circuits_wiki, drivers_wiki and constructors_wiki datasets were filtered to retain only records corresponding to entities that participated in the 2025 Formula 1 season.

The following CSV files were produced during this phase: *circuits_wiki_clean.csv*, *constructors_wiki_clean.csv*, *drivers_api_clean.csv*, *drivers_wiki_clean.csv*, *races.csv, results_api_clean.csv, sprint_api_clean.csv.*

2.2 Data Integration and Enrichment

During the correspondence investigation phase, several relationships between datasets were identified based on the presence of matching attributes. The subsequent schema integration phase allowed to combine data based on the correspondences found. More precisely:

- the drivers_api_clean and drivers_wiki_clean datasets contain complementary identifying information, namely the surname attribute and the full name attribute, respectively. To perform the merge, it was verified that the surname extracted from the API dataset was contained within the full name obtained through scraping. Once the correspondence was established, the driverID, date_of_birth, permanentNumber, name and surname attributes from the drivers_api_clean were transferred to the corresponding records in drivers_wiki_clean and the original driverName attribute was removed from drivers_wiki_clean as it became redundant.
- A similar integration strategy was applied to constructors. To facilitate the identification of correspondences, the constructors_api_clean and constructors_wiki_clean datasets were first sorted alphabetically by constructor name, allowing a direct row-by-row match between the two datasets. The constructor identifiers from the API dataset were then assigned to the matching records in constructors_wiki_clean, ensuring consistent linkage across datasets.
- Subsequently, the sprints_api_clean file was concatenated with the results_api_clean file, as they share the same structure. To distinguish between main race results and sprint race results, a binary column named “isSprint” was introduced.
- The relationship between the races dataset and result dataset was already defined through the round attribute; therefore, no additional integration steps were required in this case.
- Finally, the circuits_wiki_clean dataset was connected to the races dataset using the correspondence between the columns GrandPrix and raceName, enabling the association of each race with its respective circuit. Subsequently, the circuitID column from the races dataset was propagated to circuits_wiki based on the previously identified correspondence. As a result, the GrandPrix column was removed from circuits_wiki_clean, since this information can already be found from the races dataset.

Following these operations, each dataset contains complete information and is linked to at least one other dataset through one or more shared attributes. As a consequence all the relevant data collected are stored into races.csv and the CSV files produced in this phase: *results.csv*, *drivers.csv*, *constructors.csv*, *circuits.csv.*

The integration process was fully automated. Manual inspection was performed only to verify the correctness of the merge. No missing identifiers were found, corresponding to a total match and the absence of integration errors. This result was made possible by the data cleaning and preprocessing operations, which ensured that the datasets were complete, consistent and non-redundant.

2.3 Data Quality

This subsection evaluates the quality of the data in order to determine whether, at this stage of the pipeline, the datasets are suitable to create the dabatase.

For this project the most relevant data quality dimensions, with respect to the objectives and structure of the data, were chosen: completeness, consistency and redundancy.

- Completeness refers to the presence of all and only the relevant informations about the reality of interest and the goal of the project. While the API represents a reliable primary source and provides complete information about race events and results, it does not contain several types of data required to answer the research questions, such as circuit characteristics, historical records and aggregated career statistics for drivers and constructors. These missing aspects were addressed through web scraping and subsequent integration. Some missing values are still present; however, they are structural rather than indicative of data loss. For example, missing sprint race date and time values simply indicate that no sprint session was held for that event and missing race results for certain drivers correspond to non-classified or retired drivers. Therefore, these missing values do not compromise the analytical completeness of the dataset. The integration phase made it possible to achieve a high level of completeness with respect to the analytical goals of the project.
- Consistency refers to whether the data is recorded or collected in the same way each time. The formatting and transformation operations described in Section 2.1 ensured that equivalent attributes share the same structure, data type and representation across different sources. These operations guarantee that attributes referring to the same real-world concept are expressed consistently, enabling reliable joins and comparisons between tables. Consequently, the integrated database exhibits a coherent schema in which similar attributes are aligned across datasets, supporting accurate query execution.
- Redundancy measures the presence of duplicate or unnecessary information. At the source level, several CSV files contained overlapping information, particularly regarding driver and constructor characteristics. During the integration phase, relationships between entities were defined using unique identifiers, allowing shared information to be stored only once in dedicated tables. This normalization process eliminated most redundant attributes and ensured that descriptive data (e.g., driver personal information, constructor details) are maintained in a single location and referenced when needed.

3. DATA STORAGE

The database chosen for data storage is MongoDB, a document-oriented NoSQL database. This choice is motivated by the following main considerations:

- Compared to a relational database, MongoDB offers greater schema flexibility, which is particularly suitable for this project, as several entities contain optional or variable fields (e.g., races with or without sprint events, structural missing values). This flexibility simplifies future extensions of the database.
- Although the domain includes multiple relationships (such as driver–constructor, race–circuit and result–driver), a graph database was not selected because the main objective of the project is not to analyze the relationships, but rather to query a rich and comprehensive repository of entities characterised by a large amount of attributes. The relationships can still be effectively represented using references based on unique identifiers.
- A referenced approach was preferred because the same drivers, constructors and circuits appear in many different results. Embedding their full information inside each result document would lead to significant data duplication and more complex update operations. Using references avoids redundancy, ensures data consistency and enables efficient analytical queries that combine information across collections.

3.1 Database

The database *f1_2025* was populated by importing the cleaned and integrated CSV files produced during the previous phases. Each CSV file was read into a pandas DataFrame and the data was converted into a list of dictionaries for insertion as documents into the corresponding MongoDB collections. Before inserting new documents, the collections were cleared to ensure no outdated data remained.

The database *f1_2025* contains five collections: drivers, constructors, circuits, races and results. Collections are conceptually linked through shared identifiers, enabling efficient cross-collection queries:

- driverId links results with drivers.
- constructorId links results with constructors.
- circuitId links races with circuits.
- round links races with results.

This structure allows queries that combine information from multiple collections, providing a complete view of the 2025 Formula 1 season.

3.2 Queries

The following are the results of queries designed to answer research questions aimed at uncovering correlations and insights regarding the performances of drivers in the 2025 Formula 1 season. The queries take advantage of the database's integrated design, which combines multiple data sources, providing a complete view of drivers, races, circuits and season results.

Research question 1: Are there performance differences between races held on street circuits versus race circuits?

To answer this question, we calculated the percentage of DNF (Did Not Finished) by drivers out of the total races, grouped by circuit type. The query results are:

```
DNF Street vs Permanent Circuits
Street circuit - DNF: 26 / 180 (14.44%)
Race circuit - DNF: 46 / 419 (10.98%)
```

By integrating information about the circuits, particularly their type, it was possible to evaluate the relationship between driver performance in the 2025 season and the type of circuit. The results indicate that street circuits have a higher rate of DNF compared to permanent circuits. This could be due to the greater technical complexity and narrower layout of street circuits, which increase the likelihood of incidents and retirements.

Research question 2: Does grid position significantly influence a driver's final race position?

To answer this, a query was built to calculate the average and the average absolute difference between grid position and final race position for each driver, grouped by grid position. The query results are:

```
PERFORMANCE BY GRID POSITION
Grid 1: Avg Gain/Loss=-1.33, Avg Abs Change=1.33 (n=30 races)
Grid 2: Avg Gain/Loss=-1.43, Avg Abs Change=1.77 (n=30 races)
Grid 3: Avg Gain/Loss=-2.00, Avg Abs Change=3.00 (n=30 races)
Grid 4: Avg Gain/Loss=-1.07, Avg Abs Change=1.93 (n=30 races)
Grid 5: Avg Gain/Loss=-1.30, Avg Abs Change=2.50 (n=30 races)
Grid 6: Avg Gain/Loss=-3.67, Avg Abs Change=4.47 (n=30 races)
Grid 7: Avg Gain/Loss=-1.27, Avg Abs Change=3.13 (n=30 races)
Grid 8: Avg Gain/Loss=-2.00, Avg Abs Change=3.67 (n=30 races)
Grid 9: Avg Gain/Loss=-2.60, Avg Abs Change=4.13 (n=30 races)
Grid 10: Avg Gain/Loss=-0.40, Avg Abs Change=2.60 (n=30 races)
Grid 11: Avg Gain/Loss=-0.93, Avg Abs Change=3.80 (n=30 races)
Grid 12: Avg Gain/Loss=-1.07, Avg Abs Change=2.87 (n=30 races)
Grid 13: Avg Gain/Loss=0.00, Avg Abs Change=3.53 (n=30 races)
Grid 14: Avg Gain/Loss=-0.47, Avg Abs Change=3.33 (n=30 races)
Grid 15: Avg Gain/Loss=1.57, Avg Abs Change=3.23 (n=30 races)
Grid 16: Avg Gain/Loss=1.10, Avg Abs Change=2.97 (n=30 races)
Grid 17: Avg Gain/Loss=2.63, Avg Abs Change=3.30 (n=30 races)
Grid 18: Avg Gain/Loss=3.77, Avg Abs Change=4.03 (n=30 races)
Grid 19: Avg Gain/Loss=5.00, Avg Abs Change=5.07 (n=30 races)
Grid 20: Avg Gain/Loss=5.66, Avg Abs Change=5.66 (n=29 races)
```

The analysis shows how much drivers gained or lost positions relative to their grid position. Dividing the grid into three bands highlights interesting trends:

- Top positions (1–5): Drivers typically change at most 2 positions, generally losing a few.
- Midfield (6–12): Larger changes are observed, with both gains and losses.
- Back positions (13–20): Significant position changes are possible, with averages up to 5 positions gained or lost.

This demonstrates that grid position does influence final race outcomes, particularly for drivers starting at the front, whose positions tend to remain stable, while drivers starting further back have greater potential for movement, either positively or negatively.

Research question 3: What are the main characteristics and 2025 performance metrics of a driver?

A query was created to summarise the most significant information available in the database for a driver, covering personal data, race results and historical performance. The query results are:

```
DRIVER SUMMARY:
Name            : Charles
Surname         : Leclerc
Date of Birth   : 1997-10-16
Nationality     : Monaco
Team (Last)     : Ferrari
Seasons         : 8

Latest Season
Total races        : 30
Street circuits    : 8
Race circuits      : 16
Points             : 242
Wins               : 0
Podiums            : 7
Average position   : 7.13

Overall performances
Wins           : 8
Podiums        : 50
Poles          : 27
Total points   : 1672.0
Championships : 0
```

Thanks to the database's integrated structure, a single query, implemented through join operations. allows a user to retrieve comprehensive information about a driver, spanning personal details, career history and both overall and season-specific performance metrics.

As the results show, the database's design and integration of multiple sources enable precise answers to the research questions. Each query is able to extract insights that would not have been possible by consulting only the original sources individually, as those sources lacked the comprehensive information that the final database provides. This demonstrates the database's effectiveness in supporting detailed performance analyses and data-driven insights in Formula 1.

CONCLUSIONS

The main objective of this project was to create and implement a comprehensive database representing the 2025 Formula 1 season. Starting from heterogeneous datasets obtained via Web API and Web Scraping, data cleaning, integration and quality verification were applied to ensure consistency and reliability.

The resulting database, *f1_2025*, integrates information about drivers, constructors, races, circuits and results, with collections linked through shared identifiers. This structure allows queries that combine information from multiple collections and supports the exploration of relationships between drivers, teams, circuits, races and outcomes throughout the 2025 season.

The research questions defined at the beginning of the project were successfully answered, demonstrating that the integrated database is able to support meaningful analytical queries regarding performance differences across circuit types, the influence of grid position on race outcomes and detailed driver performance profiles.

Overall, several challenges were encountered, mainly related to the fact that data from different sources could include incomplete or inconsistent records and missing values had to be properly handled. These challenges were effectively addressed, resulting in a database that is fully functional and reliable.

The final database accurately represents the 2025 season and it is suitable for a variety of applications, such as performance analysis and statistics reporting.

Future developments could include the integration of additional seasons to enable historical comparisons and it could be keep up-to-date with ongoing Formula 1 events. Furthermore, the database could serve ad a foundation for predictive models.